

Laboratory Manual

Subject: CL 242 – Surveying

Semester: 3rd



Chandubhai S Patel Institute of Technology

Department of Civil Engineering

Academic Year: 2018-19



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CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY



Name of the Student:_____

ID No:_____ **Class:**_____

Division:_____ **Laboratory Batch:**_____

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CERTIFICATE

*This is to certify that Mr. /Ms. _____
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work in _____ for
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Surveying (CL 242)

Date:

EXPERIMENT NO. 1 SIGNS AND SYMBOLS

OBJECTIVE: Introduction to Surveying

THEORETICAL BACKGROUND:

Definition:

Surveying is an art of determining relative positions on, above or beneath the surface of earth by measuring the horizontal distances and angles for determination of any point on ground.

Levelling is an art of determining the difference of elevations or levels of various points on the surface of the earth. Levelling deals with distances in a vertical plane.

Object of Surveying & Levelling:

The object of surveying is to prepare a map / plan, with suitable scale, to show the relative positions of the objects on the surface of the earth.

- (i) Map
- (ii) Plan

The object of levelling is to determine the undulations of the earth's surface for topographic mapping. Levelling is used to design of highways, railways, canals, sewer etc.

Primary Divisions of Surveying:

- (i) Plane Surveying
- (ii) Geodetic Surveying

Necessity of levelling:

Levelling is essential for the layout of construction projects, for locating the excavation levels, and the control of various elevations in buildings, bridges, dams etc.

The drainage characteristics of area can be obtained by levelling.

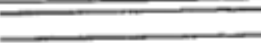
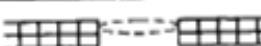
Levelling is used to determine the catchment area, volume of the reservoir and area of submerged by a reservoir.

Levelling is used to determined the volume of earthwork for roads, railways etc.

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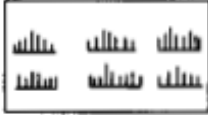
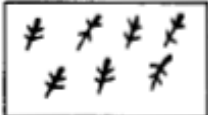
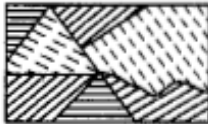
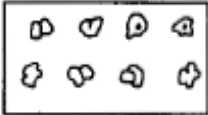
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SIGNS & SYMBOLS

Object	Symbol	Colour
Single Track		Black
Double Track		Black
Pucca Road		Vermillion Red
Kutcha Road		Vermillion Red
Footpath		Vermillion Red
Fencing Post and Rail Fence		Vermillion Red
Tunnel (Road or Rail Road)		Black
Canals and Ditches		Black
Aqueducts and Water Pipes		Black
Dam		Black
Bridge		Black
Stream		Prussian Blue
Pond		Prussian Blue
Falls and Rapids		Black
Road in Embankment		Indian Blue Ink and Road in Vermillion Red
Road in Cutting		Indian Blue Ink and Road in Vermillion Red

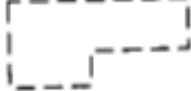
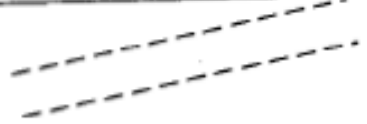
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Garden		Indian Black Ink Boundary Chain Dotted and Hopper's Green Wash
Marshy Ground		Indian Black Ink Boundary Chain Dotted and Hooper's Green Wash
Jungle		Indian Black Ink Boundary Chain Dotted and Hooper's Green Wash
Cultivated Land		Green
Orchard		Green
Fence of any kind (or Board Fence)	-----	Black
Barbed Wire Fence	-x-x-x-x-x-x-x-	Black
Smooth Wire Fence	-o-o-o-o-	Black
Rail Fence	~~~~~	Black
Hedge Fence	~~~~~	Green
Stone Fence	~~~~~	Black
Telegraph or Telephone Line	T T T T T T	Black
Power Line	-----	Black
Wall	=====	Vermillion Red
Gate		Vermillion Red

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Wall and Gate		Vermillion Red
Building (Large Scale)		Black
Building (<i>Kutcha</i>)		Umber
Huts		Yellow
Temple		Crimson lake
Church		Crimson lake
Mosque		Crimson lake
Benchmark		Black
Open well		Purssina blue
Tube well		Black
Footpath		Black
Metalled road		Burnt sienna
Unmetalled road		Burnt sienna

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Surveying (CL 242)

Date:

EXPERIMENT NO. 2 PROFILE LEVELLING

OBJECTIVE: To determine the elevations of the ground surface along the centre line.

THEORETICAL BACKGROUND:

Definition:

Profile levelling is the process of levelling along a fixed line to determine the elevations of the ground surface along the line.

Profile levelling is also known as longitudinal sectioning.

Use of profile levelling:

Profile leveling is a method of surveying that has been carried out along the central line of a track of land on which a linear engineering work is to be constructed

The fixed line is generally the centre line of a highway, railway, canal, sewer or any other such utility.

The fixed line need not be a single straight line; it may consist of different straight reaches connected by curves.

The ground profile is the section of the ground obtained when a vertical plane cuts the ground surface. The ground profile shows the elevations of the ground along the section.

The profile levelling is done to determine the undulations of the ground surface.

The ground profiles are used for the study of the relationship between the existing ground levels and the levels of the proposed route.

The gradient line is drawn and the formation levels at various points are determined and the amount of cutting and filling can be computed.

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1. In carrying out profile leveling, a level is placed at a convenient location (say I1) not necessarily along the line of observation.
2. The instrument is to be positioned in such a way that first backsight can be taken clearly on a B.M.
3. Then, observations are taken at regular intervals (say at 1, 2, 3, 4) along the central line and foresight to a properly selected turning point (say TP1).
4. The instrument is then re-positioned to some other convenient location (say I2).
5. After proper adjustment of the instrument, observations are started from TP1 and then at regular intervals (say at 5, 6 etc) terminating at another turning point, say TP2 .
6. Staff readings are also taken at salient points where marked changes in slope occur, such as that at X.
7. The distances as well as direction of lines are also measured.

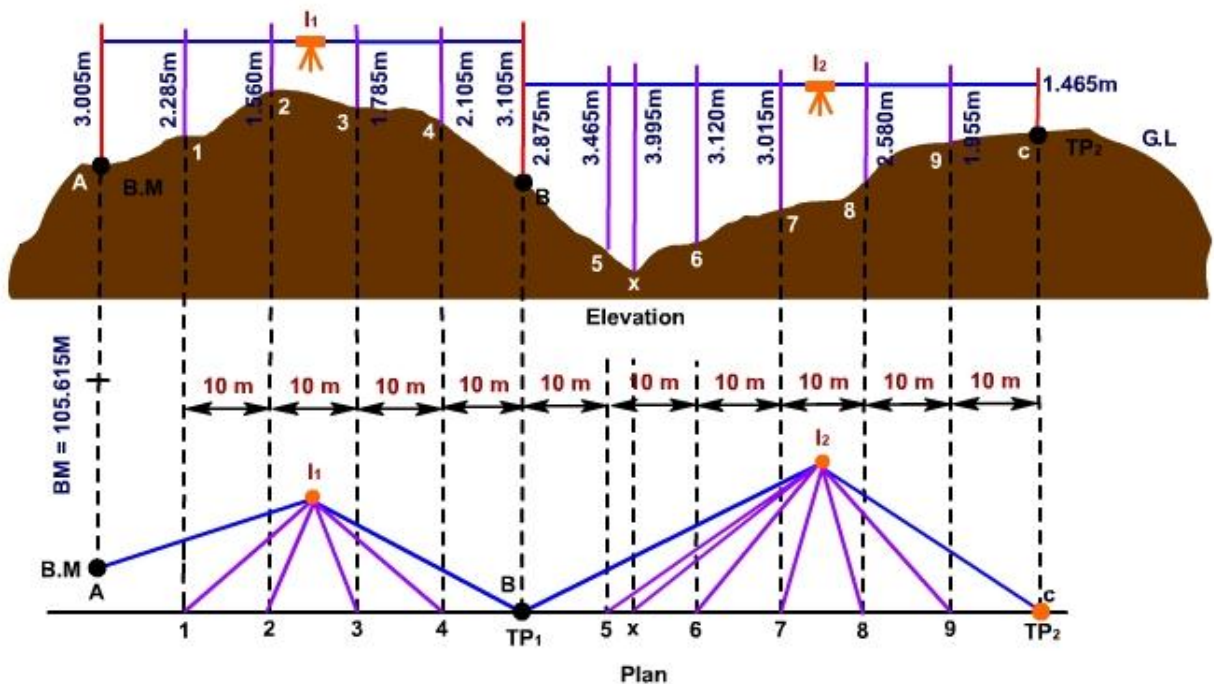


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FIELD WORK

1. The notes of profile leveling are recorded in a level note book where backsights, intermediate sights and foresights are placed in independent columns.
2. The distances of points as well as direction of lines are also noted in separate columns.
3. Instrument position at I_1 , the first backsight (B.S) is taken at B.M and the reading of 3.005m is placed in its column in the row of station A.
4. Then, intermediate sights 2.285m, 1.560m, 1.785m, 2.105m respectively at stakes 1,2,3,4 are placed in the corresponding row.
5. The first foresight 3.105m taken at station B is placed in its row.



6. From changed instrument location I_2 , a backsight 2.875m is taken at B and it is entered in the B.S. column in the row of B. Thus, at station B, both backsight and foresight readings are entered.
7. The intermediate sights 3.465m, 3.955m, 3.120m, 3.015m, 2.580m, 1.955m respectively at stakes 5, X, 6, 7, 8, 9 are placed in their corresponding row.
8. The foresight 1.465 m taken at station C is placed in its row.

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CALCULATION OF REDUCED LEVEL

Reduction of levels can be done either by height of instrument method or by rise and fall method, computations have been carried out by both the methods and subsequently their checks are done.

Pegs	Distance(m)	Direction	Staff Reading (m)			Difference in Elevation (m)		H.I (m)	R.L(m)	Remarks
			B.S	I.S	F.S	Rise	Fall			
		Σ								

$$\Sigma \text{ B.S.} - \Sigma \text{ F.S.} =$$

$$\Sigma \text{ Rise} - \Sigma \text{ Fall} =$$

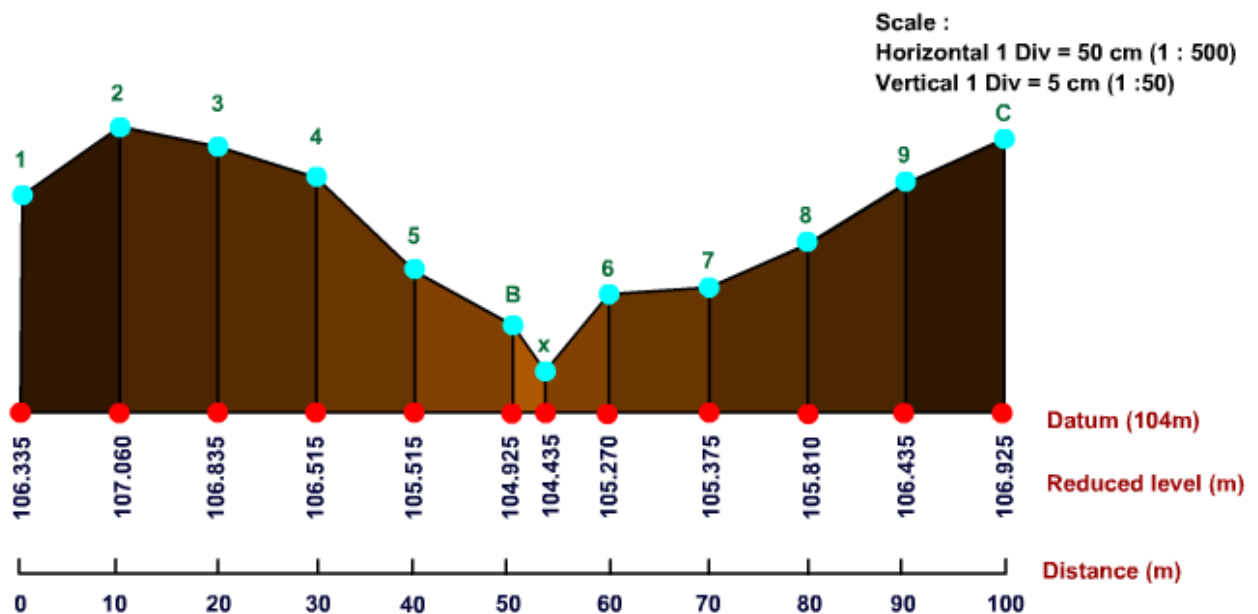
$$\text{Last R.L.} - \text{First R.L.} =$$

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PLOTTING OF PROFILE

1. Plotting of profile leveling provides a graphical representation of the ground points on a longitudinal section along the alignment.
2. It is being used to determine the depth of cutting or filling on the proposed gradient (for highways, railways, canals, etc.), to study grade crossing problems, to select appropriate grade, to locate depth of sewer, tunnels etc.
3. In this, a datum line is drawn along which distance of the stakes are marked and reduced levels are plotted along vertical lines drawn on the marked points.
4. Segmented straight lines joining the reduced level points represent the longitudinal profile of the ground surface.
5. Profile is generally drawn so that the vertical scale is much larger than the horizontal scale in order to accentuate the differences of elevations.



PROFILE OF LONGITUDINAL SECTION

In this, the datum and ground lines are drawn in black and the ordinates in gray. The value of the datum line is given and the reduced levels are written against ordinates.

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Surveying (CL 242)

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EXPERIMENT NO. 3 THEODOLITE SURVEY

OBJECTIVE: Traverse Survey with Theodolite

THEORETICAL BACKGROUND:

Definition:

The theodolite is the most precise instrument designed for measurement of horizontal and vertical angles.

Use of theodolite:

It has wide applicability in surveying such as laying off horizontal angles, locating points on the line, prolonging survey lines, establishing grade, determining difference in elevation, setting out curves etc. It is known as the universal instrument of surveying.

Types of theodolite:

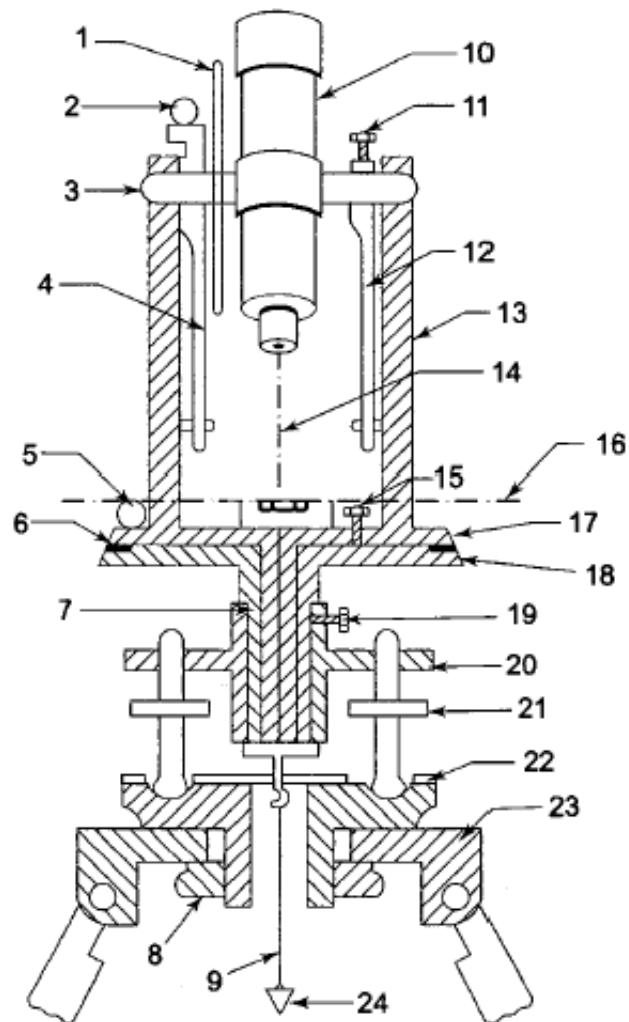
- (i) Transit theodolite : Telescope can be revolved 180° about its horizontal axis
- (ii) Non- transit theodolite: Telescope cannot be revolved 180° about its horizontal axis

COMPONENTS OF THEODOLITE:

- (i) Telescope
- (ii) Vertical Circle
- (iii) T- frame or Index bar
- (iv) Lower plate (Scale plate)
- (v) Upper plate (Vernier plate)
- (vi) Level tubes
- (vii) Tripod
- (viii) Plumb bob

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- | | |
|------------------------------------|--------------------------------------|
| 1. Vertical circle | 2. Altitude bubble |
| 3. Horizontal axes | 4. Vernier arm |
| 5. Plate bubble | 6. Graduated arc |
| 7. Levelling head | 8. Clamping nut |
| 9. Vertical axis | 10. Telescope |
| 11. Vertical circle clamping screw | 12. Arm of the vertical circle clamp |
| 13. Standard | 14. Line of sight |
| 15. Upper plate clamping screw | 16. Axis of plate bubble |
| 17. Upper plate | 18. Lower plate |
| 19. Lower plate clamping screw | 20. Tribrach |
| 21. Foot screw | 22. Trivet |
| 23. Tripod top | 24. Plumb bob |

Transit theodolite

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PROCEDURE TO MEASURE HORIZONTAL ANGLE WITH THEODOLITE:

Station	Object	Angle	Observation	Reading on Vernier		Angle on Vernier		Mean Angle on Vernier	Mean Angle of observation	Remark
				A	B	A	B			
1	2	3	4	5	6	7	8	9	10	11
O	A	AOB	Face Left	0°0'0"	180°0'0"					
	B									
O	A	AOB	Face Right	0°0'0"	180°0'0"					
	B									

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PROCEDURE TO MEASURE VERTICAL ANGLE WITH THEODOLITE:

Station	Object	Angle	Observation	Reading on Vernier		Angle on Vernier		Mean Angle on Vernier	Mean Angle of observation	Remark
				C	D	C	D			
1	2	3	4	5	6	7	8	9	10	11
O	Hor. A	AOC	Face Left	0°0'0''	0°0'0''					
	Hori. A		Face Right	0°0'0''	0°0'0''					
O	Hor. B	BOC	Face Left	0°0'0''	0°0'0''					
	Hori. B		Face Right	0°0'0''	0°0'0''					

Calculation :

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**EXPERIMENT NO. 4
PLANE TABLE SURVEY**

OBJECTIVE: To Locate the given building by plane table surveying.

THEORETICAL BACKGROUND:

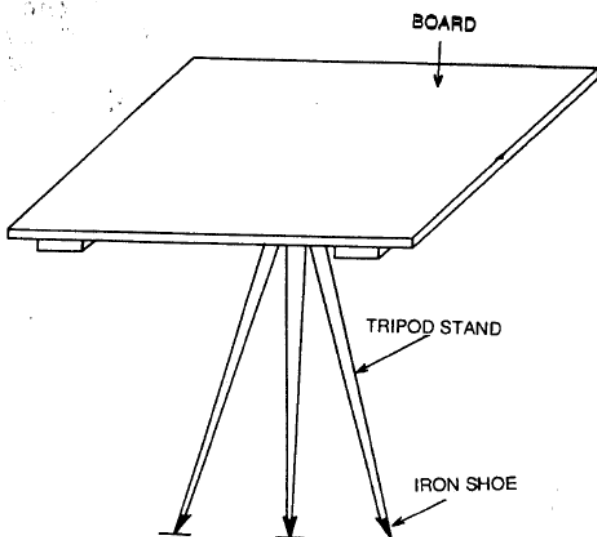
Principle:

The principle of plane tabling is parallelism, means that the ray drawn from station to objects on the paper are parallel to the lines from the stations to be objects on the ground.

Relative positions of the objects on the ground are represented by their plotted positions on the paper and lie on the respective rays.

Accessories of Plane Table:

The Plane table with tripod, Alidade, Trough compass, Sprit level , Plumbing fork or Plumb bob, Tape ,chain, pegs, ranging rods, wooden mallet etc.



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Setting Up the Plane Table

- (i) Fixing the Plane Table
- (ii) Centering the Plane Table
- (iii) Levelling the Plane Table

Orientation :

Method of setting up the plane table at each of successive stations parallel to the position it occupied at starting station is called as orientation.

- (i) Orientation by magnetic Needle

The method is suitable when local attraction is not suspected in the area.

- (ii) Orientation by back sighting

This method is accurate and is always preferred.

Traversing :

- (i) Radiation Method of Traversing.
- (ii) Intersection Method of Traversing.
- (iii) Plane Table Traversing
- (iv) Resection
- (v)

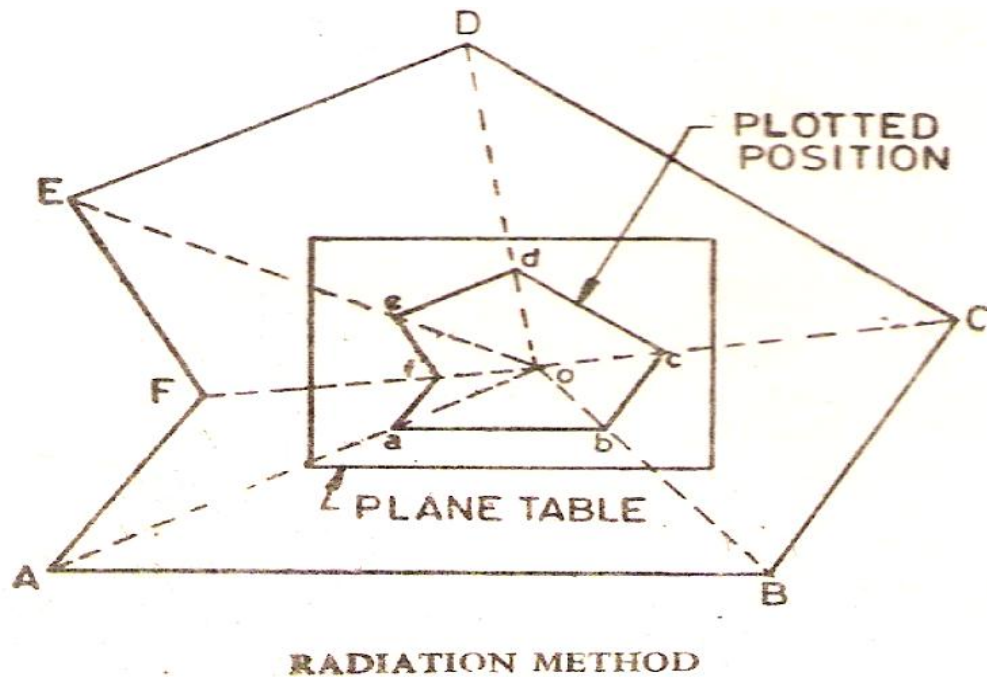
Radiation Method:-When from a single set of plane table on instrument station different details are located on the sheet, the method is known as radiation method.

In this method the rays are drawn from the instrument station to the point to be located, then the distances are measured from the instruments station to the point and the position of the each point is plotted on the sheet using a suitable scale.

The method is most suited for surveying small areas which can be controlled by single setting. It can also be used in combination with other method. This method can be applied for locating distant points if the distances are obtained tacheometrically with the help of the telescope alidade.

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PROCEDURE:

- 1) Select the position of the table where it is to be set so that all the points to be located are visible from it. Let 'O' be the position of such a point on the ground.
- 2) Set the plane table over this point and level it. Draw the North line in the top corner of sheet by means of trough compass at the table.

Now transfer the position of the point 'O' on the ground to the sheet by means of a plumbing fork. The point 'O' will represent point 'o' will represent point 'O' on the ground.

With the alidade touching the point 'o' (may be represented by fixing a pin), sight the point A in the field. Draw the ray along the fiducial edge. Measure the distance of this point from the instrument station by means of tape and plot the point 'a' corresponding to point 'A' in the field to scale in the sheet.

Similarly sight other points such as B, C, D, E etc. and measure their distances from the instrument station. Plot them to scale to get their position on the sheet such as b,c,d etc. on the sheet.

Intersection method:-When the location of an object is obtained on the sheet of paper by the intersection of the rays drawn after sighting at the object from two plane table stations (previously plotted), it is called intersection method.

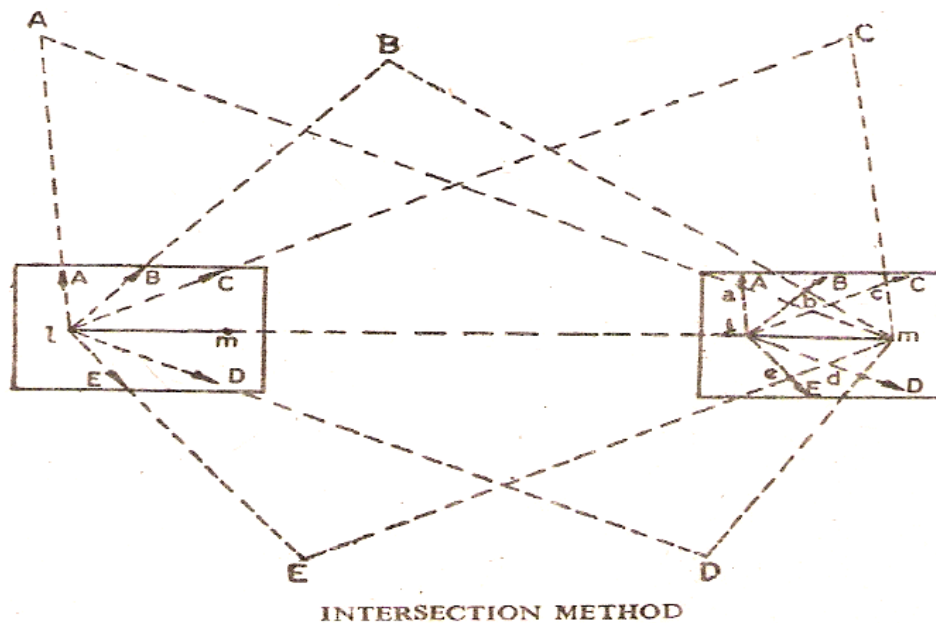
The method is suitable when the distance between the point and the instrument station is either too large or cannot be measured accurately due to some field conditions as in case of mountainous country. It is also employed for filling up details, locating distant and inaccessible

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object, locating the broken boundaries as in the case of rivers etc. The method can also be used for checking of plotted points.

The line joining the two instrument stations is known as the base line. No linear measurement other than the base line is made.



PROCEDURE:

1) Select two points L and M in such a way so that all the points to be plotted are visible from them. Now set the table at station, point L in such a position so that the sheet should cover all the points. Level the table and clamp it.

Draw the north line in the top corner of sheet by means of trough compass

Now transfer the position of station point L on the sheet as T with the help of plumbing fork so that it is vertically above the instrument station.

With the alidade pivoted about T sight the ranging rod fixed at station point M and draw the line in the direction of M. Now measure the distance LM by means of the tape and cut off lm to some suitable scale along the ray drawn toward M; thus fixing the position of 'm' on the sheet corresponding to station point M on the ground. The line lm is called the base line.

5) With the alidade touching the point 'l' sight the objects in the field such as A,B,C,D,E etc. as shown in figure and draw the rays towards them. The direction of each line is marked with an arrow and a letter A,B,C,D,E etc. corresponding to above details.

Now shift the table to the station point M and approximately set it in the line with ML. Set it up so that the point 'm' is vertically above the station point 'M' and level it.

Orient the table roughly by compass, then finally by placing the alidade along ml and bisecting the ranging rod fixed at station point 'L' i.e by back sighting 'L'. Clamp the table in this position.

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With the alidade centered at m sight the same object in the field such as A, B, C, D, E etc; and draw rays. The intersection of these rays with the respective rays from l locate the object A,B,C,D,E etc; as a ,b,c,d,e etc; on the sheet

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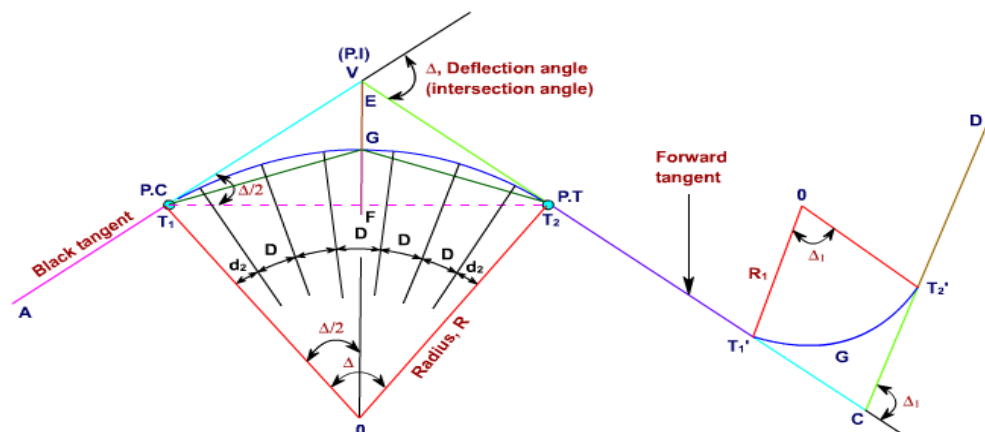
OBJECTIVE: To set out simple circular curve.

Definitions

1. Horizontal Curve
 - a. Simple curve
 - b. Compound Curve
 - c. Reverse Curve
 - d. Composite Curve
 - e. Transition Curve
2. Vertical Curve
 - a. Sag Curve
 - b. Summit Curve

- Radius of curve
- Degree of curve.

Elements of Simple curve



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Methods of Setting out curve

1. Linear method

- Offsets from long chord method
- perpendicular offsets from the tangent method
- Radial offset from the tangent method
- Successive bisection method.

2. Angular method

- Rankin's Method of deflection angle
- Two-theodolite method.
- Tacheometric method.

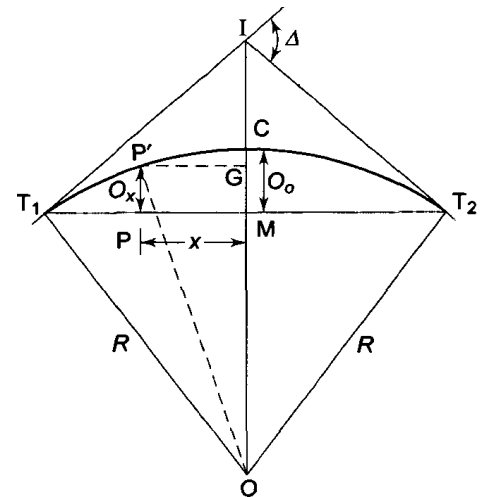
Offset from long chord method

Let it be required to lay a curve T_1CT_2 between the two intersecting straight T_1I and T_2I .

R is the radius of the curve,

O_0 the mid-ordinate,

O_x the offset at a point P at any distance x from the mid-point (M) of the Long chord.



From triangle OMT_1 :

$$\begin{aligned} OM &= \sqrt{(OT_1^2 - MT_1^2)} \\ &= \sqrt{(R^2 - (L/2)^2)} \end{aligned}$$

Now,

$$CM = OC - OM$$

or

$$O_0 = R - OM$$

or

$$O_0 = R - \sqrt{R^2 - (L/2)^2}$$

In triangle $OP'G$

$$OG = \sqrt{R^2 - x^2} \text{ and } OM = R - O_0$$

The required offset

$$PP' = OG - OM$$

Hence,

$$PP' = \sqrt{R^2 - x^2} - (R - O_0)$$

or

$$O_x = \sqrt{R^2 - x^2} - (R - O_0) \quad (\text{exact expression})$$

$$= R(1 - x^2/R^2)^{1/2} - R + O_0$$

$$= R(1 - x^2/(2R^2) + \dots) - R + O_0$$

$$= O_0 - \frac{x^2}{2R}$$

(approximate expression)

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Set A curve with following data

Two roads meet at an angle of $127^{\circ}30'$. Calculate the necessary data for setting out a curve of 15 chains radius to connect the two straight portions of the road if it is intended to set out the curve by chain and offset only. Assume 20 m chain.

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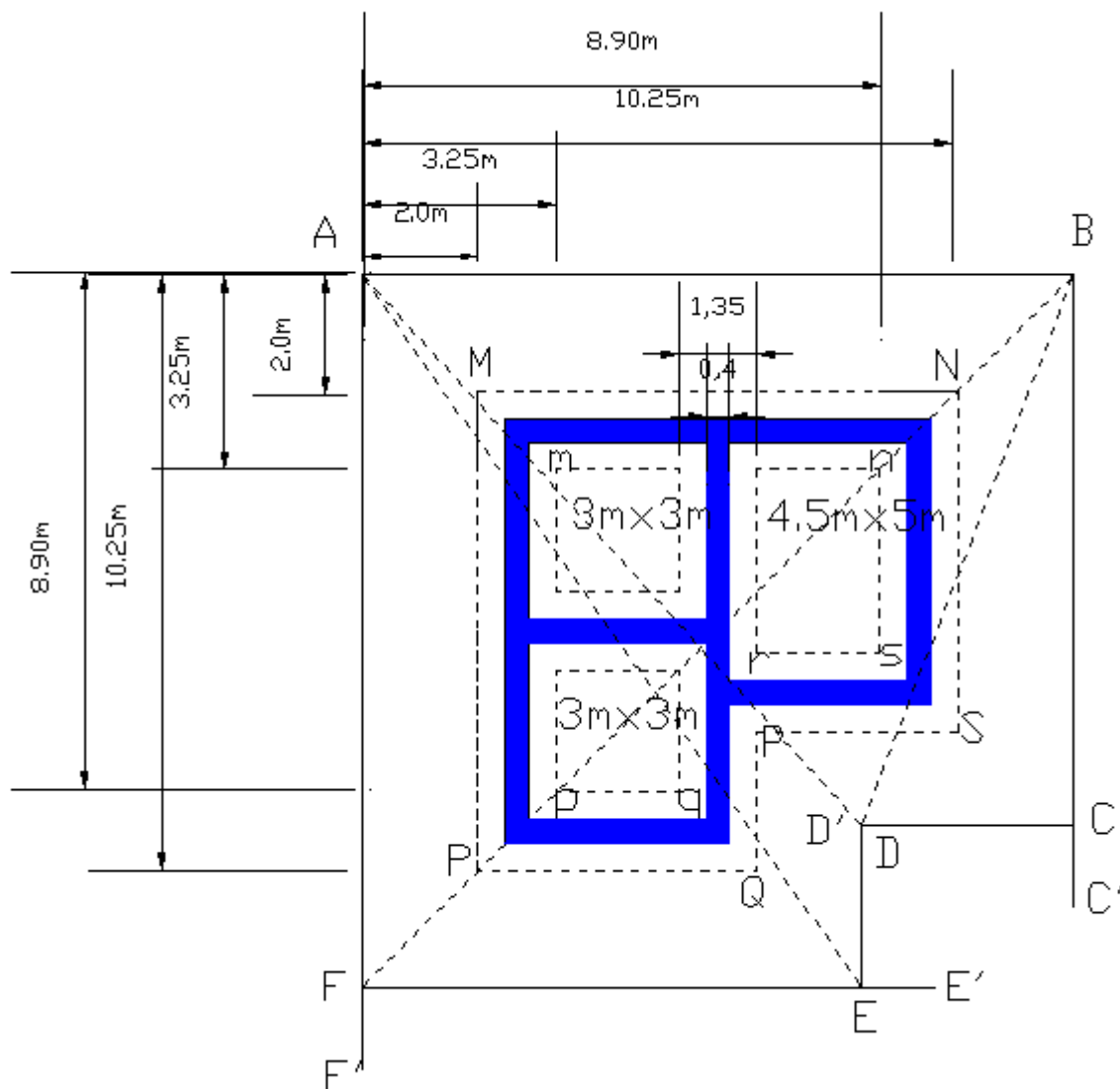
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EXPERIMENT NO.6

SETTING OUT BUILDING

OBJECTIVE: To give Layout for given plan of building.

APPARATUS: Pegs, Nails, Lime, Wooden Mallet



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THEORY: - When plans are ready for the works, the works are to be executed. To start with any structure first of all, trenches for the foundation are to be excavated. To excavate these trenches, the outline of excavation is defined on the ground, the process of defining the outlines of the excavation on the ground is known as setting out of works or lining out of works. To set any structure or work whether it may be building, culvert, pipeline or sewer line, the plan showing the width of the foundation trench, for various walls, distance of the corners from some definite line etc. is required. This plan is called foundation plan (Fig). The distances and they are with reference to lines AB and AF.

PROCEDURE:-

To start with the setting out of building, first of all a point A is fixed and then line AB is oriented in the required direction. Thus having fixed the direction of the line AB, two pegs A and B are driven at a distance of 12.25 m, apart (This distance is calculated from the plan). Wire nails are driven at the centers of the pegs. Again the distance between the wire nails is checked and which should be equal to 12.25m. A cord is stretched along AB and ends are secured to these wire nails at A and B, perpendicular AF' and BC' are set out. Perpendicular may be set with a tape by 3-4-5 method or theodolite may be used if the work is important. Along AF' and BC', point F and C are fixed at 12.80m and 10.30m, from A & B respectively. The perpendicular are then set at C and F and point D and E are fixed along CD' and FE' at a calculated distance from C and F respectively. The stakes are driven at this point C, D, E, and F and wire nails are driven at the centers of these stakes. A cord is stretched all along ABCDEF.

To check up the work, the diagonal AE, AD, BF, and BD are actually measured and these measured values should agree with their corresponding calculated lengths. Otherwise the setting out work should be repeated and stakes should be re fixed at their correct positions.

After fixing up all the pegs and stretching the cord the corners M, N, P etc. and m, n, p etc. are to be located. The point A is considered as the origin and the lines AB and AF as the axes of the co-ordinates. The co-ordinates of all the corners M, N, P etc., and m, n, p, etc. are calculated with reference to A as origin.

For example Co-ordinates of M, N, P are (2, 2), (2, 10.25) and (10.25, 2) respectively and those for m, n, p are (3.35, 3.35), (3.35, 8.90) and (8.90, 3.35) respectively. With these co-ordinates, point M, N, P, m, n, p etc., are set and pegs are driven at these points. The cord is stretched

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around the wire nails at M, N, P, Q, R, S and m, n, p, r, s indicating peripheries. The outlines of the peripheries are marked with lime spread.

Now the lime lines on the ground indicate the trenches for the various walls and the excavation may be started .if during the progress of the work, the lines marked are disturbed, it may be checked or reset with help of reference line ABCDEF.

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Surveying (CL 242)

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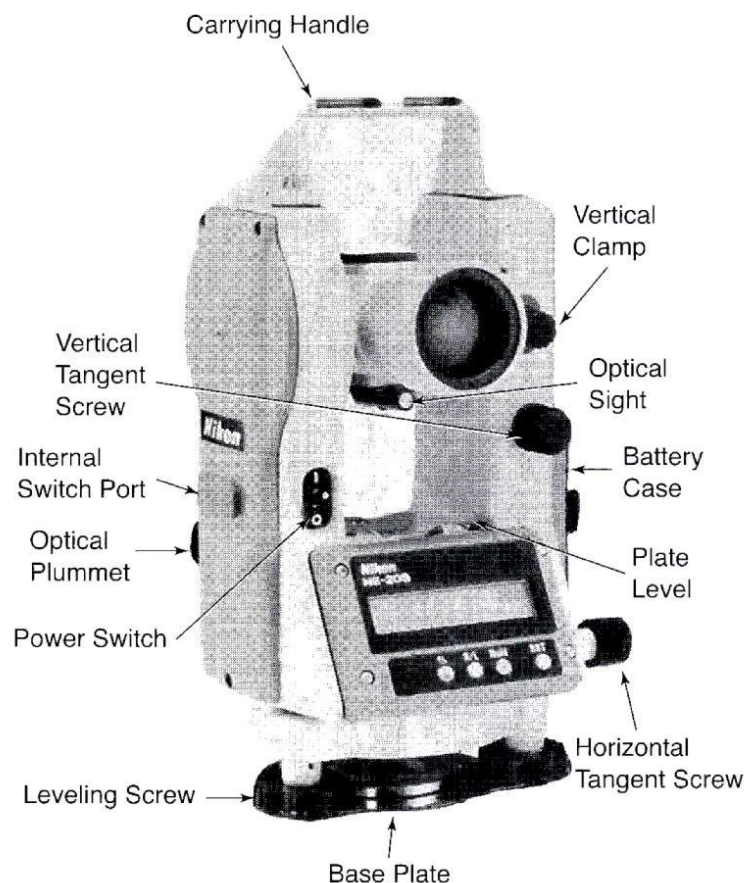
EXPERIMENT NO.7 USE OF ELECTRONIC THEODOLITE & TOTAL STATION

OBJECTIVE: Practice surveying work with electronic theodolite and total station

APPARATUS: Peg, Total station, Electronic Theodolite, Prism, Prism rod.

ELECTRONIC THEODOLITE:

- Electronic read out 1" eliminate mistakes and reading the angles.
- Precision varies from 0.5" – 20"
- Zero is set by a button.
- Repeated angle averaging.
- Replacing optical theodolite (It is less expensive to purchase and maintain).



Electronic theodolite

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TOTAL STATION

A total station is an electronic/optical instrument used in modern surveying.

The total station is an electronic theodolite (transit) integrated with an electronic distance meter (EDM) to read slope distances from the instrument to a particular point. It records all the data digitally and it used later to prepare survey map or plan. It is also used to set out the works.



Total Station

Survey with Electronic Theodolite & Total station

1. Set out simple circular curve with help of electronic theodolite.
2. Carry out traverse survey with help of Total station.

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EXPERIMENT NO.8 INDIRECT METHOD OF CONTOURING

OBJECTIVES: Draw the contour by indirect methods (grid method)

APPARATUS: Dumpy level, Leveling staff, Wooden peg, Tape, ranging rod

THEORY:

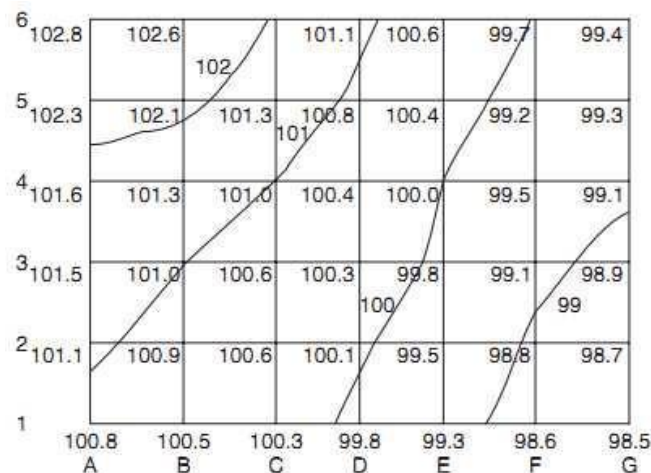
Indirect Methods

In this method, levels are taken at some selected points and their levels are reduced. Thus in this method horizontal control is established first and then the levels of those points found. After locating the points on the plan, reduced levels are marked and contour lines are interpolated between the selected points.

For selecting points anyone of the following methods may be used:

- (a) Method of squares,
- (b) Method of cross-section, or
- (c) Radial line method.

Method of Squares: In this method area is divided into a number of squares and all grid points are marked (Ref. Fig.)



Contour with Method of Square

Fig. 1

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Commonly used size of square varies from 5 m × 5 m to 20 m × 20 m. Levels of all grid points are established by levelling. Then grid square is plotted on the drawing sheet. Reduced levels of grid points marked and contour lines are drawn by interpolation.

Drawing Contours

After locating contour points smooth contour lines are drawn connecting corresponding points on a contour line. French curves may be used for drawing smooth lines

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EXPERIMENT NO.9 FIELD PROJECT ON THEODOLITE TRAVERSING

OBJECTIVE: Plot a closed traverse forming a closed polygon

APPARATUS: Theodolite, Ranging rod, Peg, Staff, Tape

THEORY:

THEODOLITE TRAVERSING:

Methods for closed traverse

- Included angle method
- Fast angle (or magnetic bearing method)

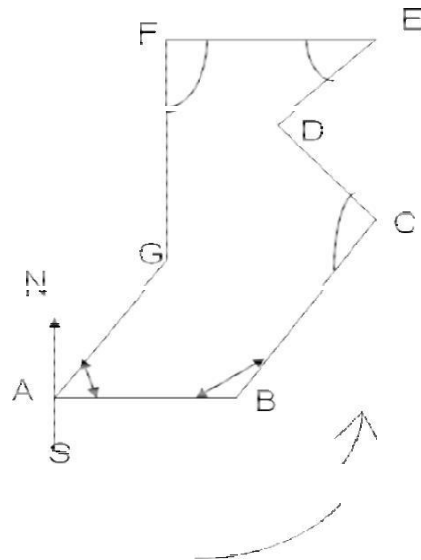
Included angle method

For running the traverse ABCDEFG Set up the theodolite at 1st station A and observed the bearing of the line AB.

Then measure the angle GAB. Shift the instrument to each of the successive station B,C etc. and measure the angles ABC,BCD etc.

Measure the line AB,BC,CD etc. and take offset to locate the required detail after this check is applied for interiors angles it is $(2n-4) \times 90^\circ$,

And for exterior angles it is $(2n+4) \times 90^\circ$ n = number of sides of the traverse



Fast angle (or magnetic bearing method)

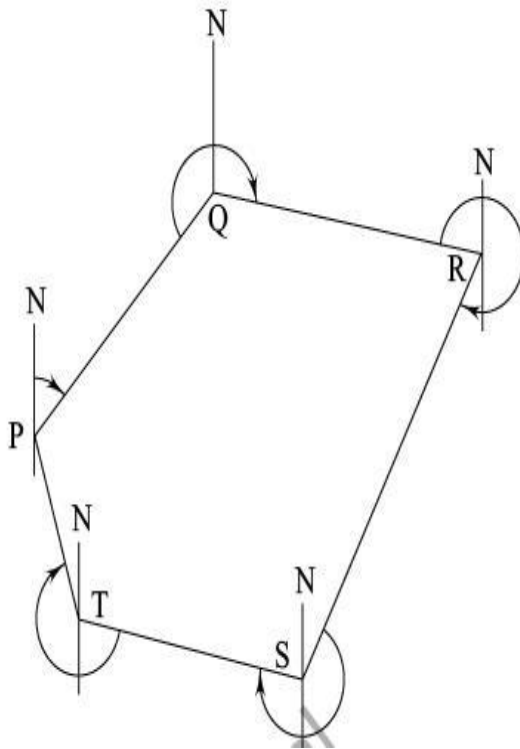
Set up and level the theodolite at station P of the traverse PQRSTP, a closed traverse.

Using the upper clamp and upper tangent screw, set vernier A to read zero.

Loosen the magnetic needle. Release the lower clamp and point the telescope in the direction of the magnetic meridian till the magnetic needle comes to rest at the zero position using the lower tangent screw the north end of the magnetic needle to read exactly zero.

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Release the upper plate and swing the instrument to bisect the signal at Q. With the upper tangent screw, bisect the station mark exactly. Read vernier A, this gives the bearing of the line PQ.

Keeping both the clamps tight, shift the instrument to Q. Set up and level the instrument.

Check the reading on vernier A. It should be the same as the magnetic bearing of the line PQ (if not, this can be corrected and the bearing value noted earlier be set on vernier A).

Release the upper clamp. Swings the instrument clockwise to bisect the station mark at R. Using upper tangent screw bisect mark R exactly. Read the vernier at A and note down the reading.

With both clamps tight, shift the instrument to R and repeat the procedure. The work is continued at all stations in a similar manner.

Observation Table Fast Angle Method

Inst. Station	Observation	Reading on Vernier		Line	Length	Bearing
		A	B			

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Observation Table Included Angle Method

Inst. Station	Object	Face	Angle	Reading on Vernier		Angle on Vernier		Mean Angle of Vernier	Mean Angle of observation	Length	WCB
				A	B	A	B				
A	E	Left	A								
	B										
A	E	Right	A								
	B										
B	A	Left	B								
	C										
B	A	Right	B								
	C										
C	B	Left	C								
	D										
C	B	Right	C								
	D										
D	C	Left	D								
	E										
D	C	Right	D								
	E										
E	D	Left	E								
	A										
E	D	Right	E								
	A										

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Exercise

Apply appropriate correction to balance the traverse and Prepare Gale's Traverse Table.

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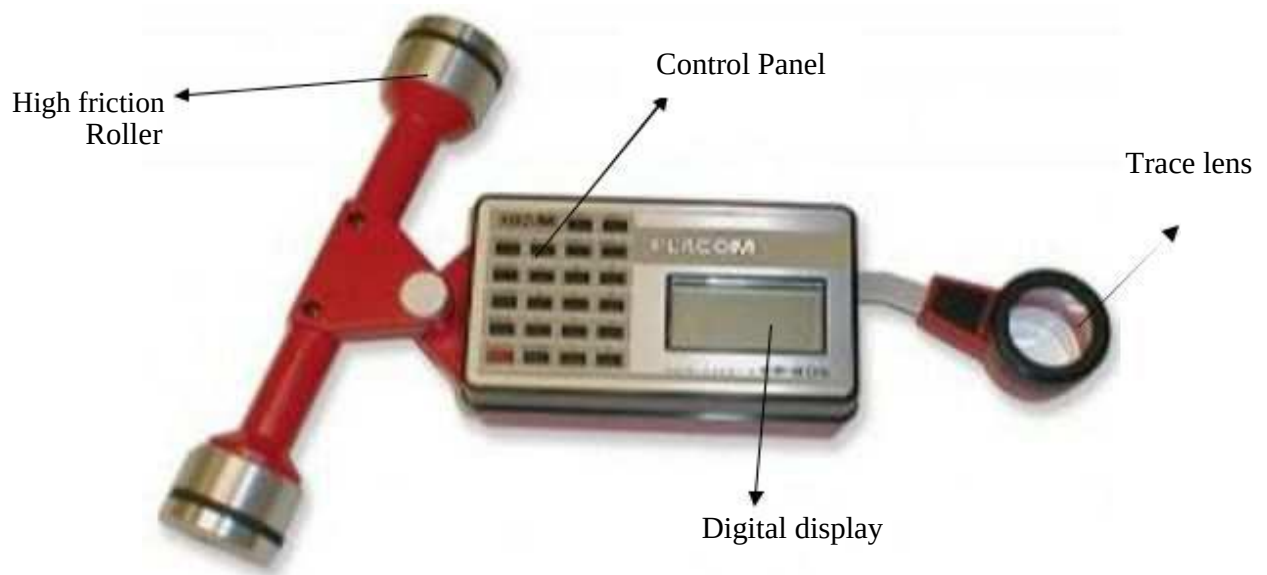
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EXPERIMENT NO.10
MEASURE AREA OF CLOSE TRAVERSE

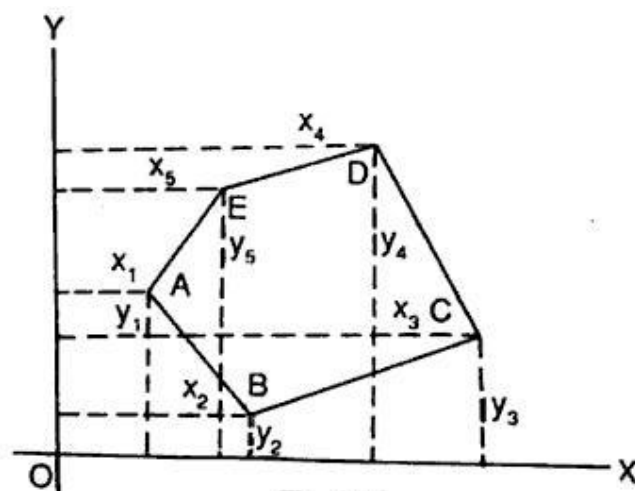
OBJECTIVE: Measure Area of close traverse with digital planimeter and analytic method.

APPARATUS: Digital Planimeter

THEORY:



Area By method of co-ordinates



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Point	y	x	Positive product (Solid product)	Negative product (Dashed product)
1	y ₁	x ₁		
2	y ₂	x ₂	x ₁ y ₂	y ₁ x ₂
3	y ₃	x ₃	x ₂ y ₃	y ₂ x ₃
4	y ₄	x ₄	x ₃ y ₄	y ₃ x ₄
5	y ₅	x ₅	x ₄ y ₅	y ₄ x ₅
1	y ₁	x ₁	x ₅ y ₁	y ₅ x ₁
			Sum 1	Sum 2
$\text{Area} = \frac{1}{2} (\text{Sum1} - \text{Sum 2})$				

Exercise

Find the Area of Traverse prepared in experiment no 9 with digital Planimeter and verify answer with method of coordinates.

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